



Introduction to Statistical Methods

BITS Pilani

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ISM TEAM

Central tendency
Variability

Probability

Conditional

Total probability

Course No: AIMLCZC418

Webinar 1 : 12.05.2026

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Topics - Webinar



- Descriptive Statistics
 - ❖ Measures of Central Tendency ✓
 - ❖ Measures of Variability ✓
- Probability
 - ❖ Introduction and Basics ✓
 - ❖ Conditional probability ✓

Example:

Median → Middle



To find the median, when the dataset has an even number of values:

8, 10, 13, 15, 17, 21

Formula: For even n :

$$\text{Median} = \frac{\left(\frac{n}{2}\right)\text{th value} + \left(\frac{n}{2} + 1\right)\text{th value}}{2} \quad \checkmark$$

For odd n :

$$\text{Median} = \left(\frac{n + 1}{2}\right)\text{th value}$$

The middle positions are:

$$\frac{n}{2} = \frac{6}{2} = 3 \checkmark$$

and

$$\frac{n}{2} + 1 = 4 \checkmark$$

So the **3rd** and **4th** values are **13** and **15**.

◆ Find the Median

$$\text{Median} = \frac{13 + 15}{2} = \frac{28}{2} = 14 \checkmark$$

Mode, Bimodal, and Multimodal



Repeated more no of times

A given set of data may have one or more than one Mode. A set of numbers with one Mode is unimodal, a set of numbers having two Modes is bimodal, a set of numbers having three Modes is trimodal, and any set of numbers having four or more than four Modes is known as multimodal.

Bimodal Mode – A set of data including two modes is identified as a bimodal model. This indicates that there are two data values that possess the highest frequencies. For example, the mode of data set B = { 8, 12, 12, 14, 15, 19, 17, 19} is 12 and 19 as both 12 and 19 are repeated twice in the given set.

No mode:

If no number in a set of numbers occurs more than once, that set has no mode: 3, 6, 9, 16, 27, 37, 48.

A unimodal mode is a set of data with only one mode.

A bimodal mode is a set of data that has two modes.

A trimodal mode is a set of data that has three modes.

$$\text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ obs} = \left(\frac{15+1}{2} \right)^{\text{th}} \text{ obs} = 8^{\text{th}}$$

$$n = 15$$

Measures of Central Tendency

1) A psychologist wrote a computer program to simulate the way a person responds to a standard IQ test. To test the program, he gave the computer 15 different forms of a popular IQ test and computed its IQ from each form.

IQ Values:

134	136	137	138	138	143	144	144	145	146	146	146
147	148	153									

$$\text{Mean} = \frac{134 + 136 + 137 + \dots + 153}{15} = \frac{2145}{15} = 143$$

$$\text{Range} = 153 - 134 = 19$$

Find the following Statistical measures:

- Mean, median, and mode
- Range, Variance and standard deviation
- The interquartile range. $\rightarrow Q_3 - Q_1$
- Identify potential outliers, if any.
- Construct and interpret a boxplot

Measures of Central Tendency

Arranging the data in ascending order:

134, 136, 137, 138, 138, 143, 144, 144, 145, 146, 146, 146, 147, 148, 153

N=15

Measure	Formula	Solution
Mean	$\mu = \frac{\sum_{i=1}^n x_i}{n}$	$\mu = 143$
Median	$p = \frac{n + 1}{2}$	Median = 144
Mode	The mode is the value or values that occur most frequently in the data set. A data set can have more than one mode, and it can also have no mode	Mode = 146

Pop. $\mu = \text{mean}$ $\bar{x} \xrightarrow{\text{sam}} \text{mean}$

Measures of Variability

Range	Range = $x_n - x_1$	Minimum = 134 Maximum = 153 Range R = 19
Variance	For a Population $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$ For a Sample $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$	Variance = 26
Standard deviation	For a Population $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$ For a Sample $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$	Standard Deviation = 5.09901951

$$\begin{aligned}
 \sigma^2 &= \frac{\sum (X - \mu)^2}{N} \\
 &= \frac{\sum (X^2 - 2\mu X + \mu^2)}{N} \\
 &= \frac{\sum X^2}{N} - \frac{2\mu \sum X}{N} + \frac{N\mu^2}{N} \\
 &= \frac{\sum X^2}{N} - 2\mu^2 + \mu^2 \\
 &= \frac{\sum X^2}{N} - \mu^2
 \end{aligned}$$

variance SD



Deviations from the mean & their squares:

x	$x - \bar{x}$ <u>$x - 143$</u>	$(x - \bar{x})^2$ <u>$(x - 143)^2$</u>
134	-9	81
136	-7	49
137	-6	36
138	-5	25
138	-5	25
143	0	0
144	1	1
144	1	1
145	2	4
146	3	9
146	3	9
146	3	9
147	4	16
148	5	25
153	10	100

Sum of squared deviations

$$81 + 49 + 36 + 25 + 25 + 0 + 1 + 1 + 4 + 9 + 9 + 9 + 16 + 25 + 100 = 390$$

✓ $\Sigma(x - \bar{x})^2 = 390$

Variance:

If this is a sample:

$$s^2 = \frac{390}{n-1} = \frac{390}{14} = 27.857$$

If this is a population:

$$\sigma^2 = \frac{390}{n} = \frac{390}{15} = 26$$

Standard Deviation:

Sample SD:

$$s = \sqrt{27.857} = 5.28$$

Population SD:

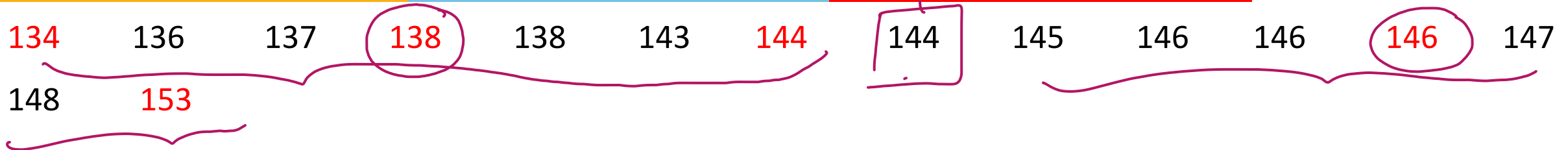
$$\sigma = \sqrt{26} = 5.10$$

$SD = \sqrt{Var}$

$$\left(\frac{15+1}{2}\right) = 8^{\text{th}}$$

25% Q_1
50% $Q_2 = \text{medi}$
75% Q_3

Measures of Central Tendency



❖ Minimum. = 134

❖ Q1 (the first quartile, or the 25% mark). 138

❖ Median. = $Q_2 = 144$

❖ Q3 (the third quartile, or the 75% mark). 146

❖ Maximum. = 153

$$\begin{aligned} IQR &= Q_3 - Q_1 \\ &= 146 - 138 \\ &= 8 \end{aligned}$$

Outliers

$$\begin{aligned} \text{Lower fence} &= Q_1 - (1.5) IQR = 138 - (1.5) 8 = 138 - 12 = \underline{126} \\ \text{Upper fence} &= Q_3 + (1.5) IQR = 146 + (1.5) 8 = 146 + 12 = \underline{158} \end{aligned}$$

Measures of Central Tendency

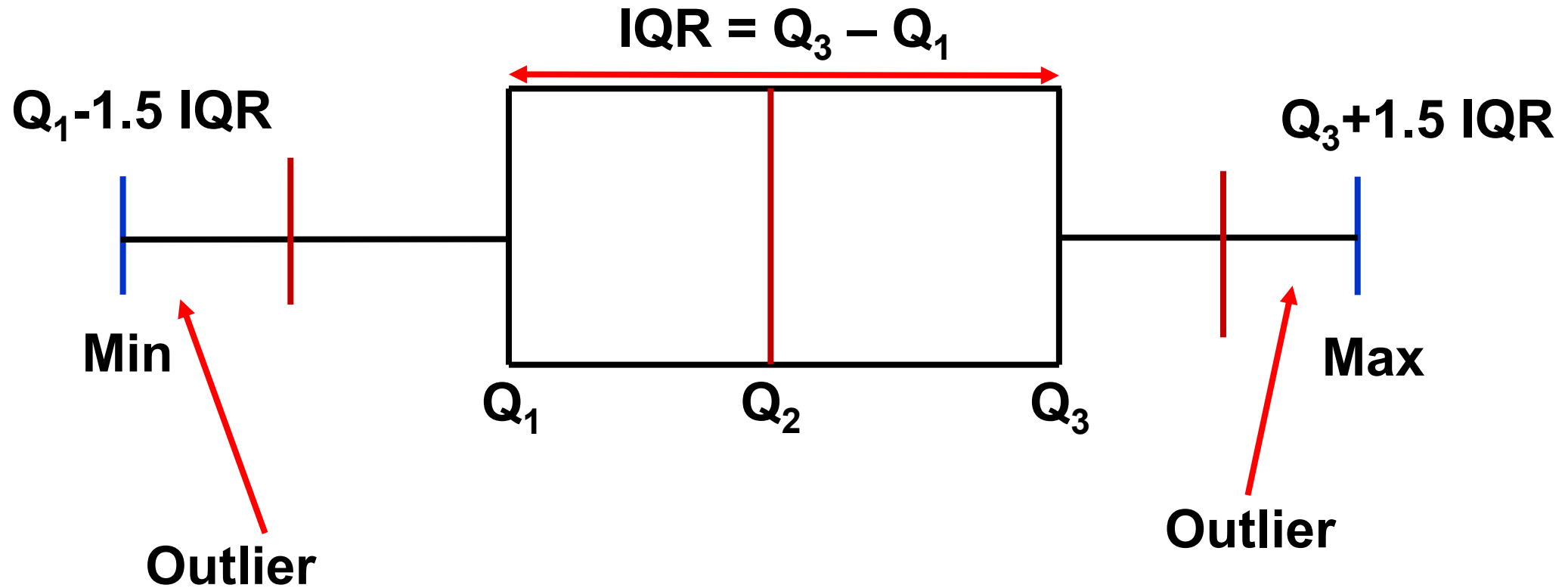
Quartiles	Quartiles separate a data set into four sections. The median is the second quartile Q_2. It divides the ordered data set into higher and lower halves. The first quartile, Q_1, is the median of the lower half not including Q_2. The third quartile, Q_3, is the median of the higher half not including Q_2.	Quartiles: <table> <tr> <td>Q_1</td> <td>--></td> <td>138</td> </tr> <tr> <td>Q_2</td> <td>--></td> <td>144</td> </tr> <tr> <td>Q_3</td> <td>--></td> <td>146</td> </tr> </table>	Q_1	-->	138	Q_2	-->	144	Q_3	-->	146
Q_1	-->	138									
Q_2	-->	144									
Q_3	-->	146									
Interquartile range	$IQR = Q_3 - Q_1$	Interquartile Range $IQR = 8$									
Potential outliers, if any.	Upper Fence = $Q_3 + 1.5 \times IQR$ Lower Fence = $Q_1 - 1.5 \times IQR$	none									

$Q_1 - 3\text{IQR}$

$Q_3 + 3\text{IQR}$



Graphical Representation: Box plot



Box-plot for the given data:



Population size: 15

Median: 144 ✓

Minimum: 134 ✓

Maximum: 153 ✓

First quartile: 138 ✓

Third quartile: 146 ✓

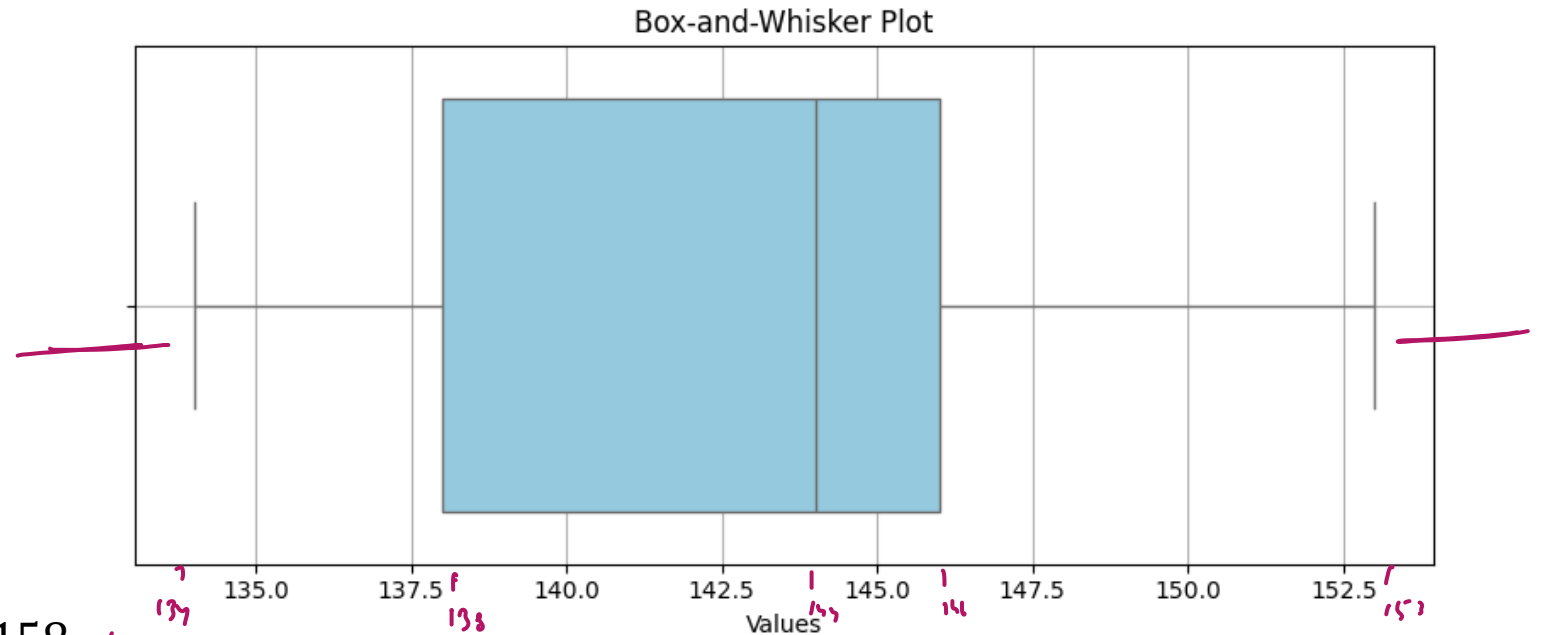
Interquartile Range: 8 ✓

Outliers: none ✓

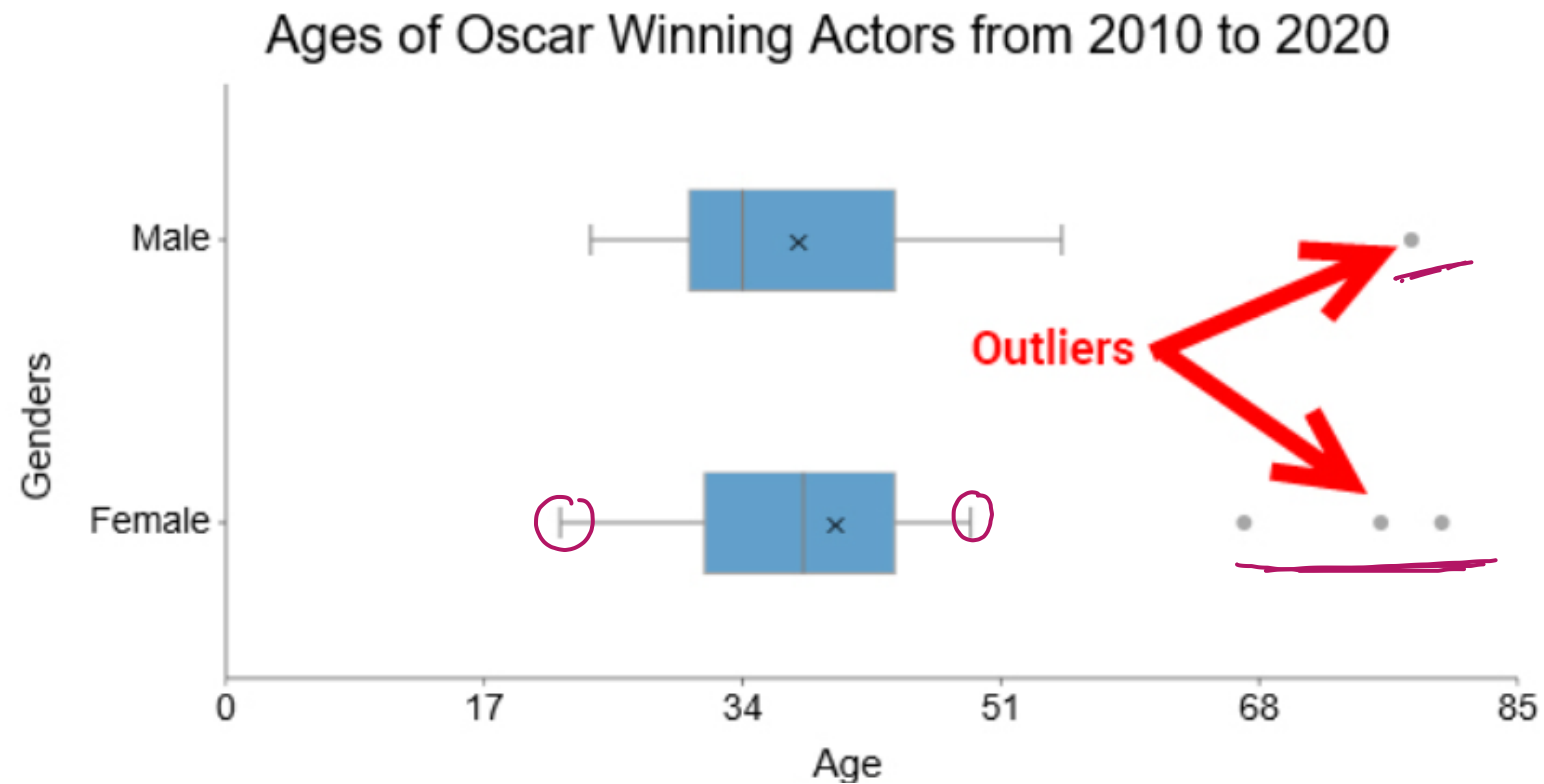
$$\text{LRL} = 138 - 1.5(8) = 138 - 12 = 126 \checkmark$$

$$\text{URL} = 146 + 1.5(8) = 146 + 12 = 158 \checkmark$$

$\begin{matrix} < 126 \\ > 158 \end{matrix} \}$ outliers



Real-time Example:



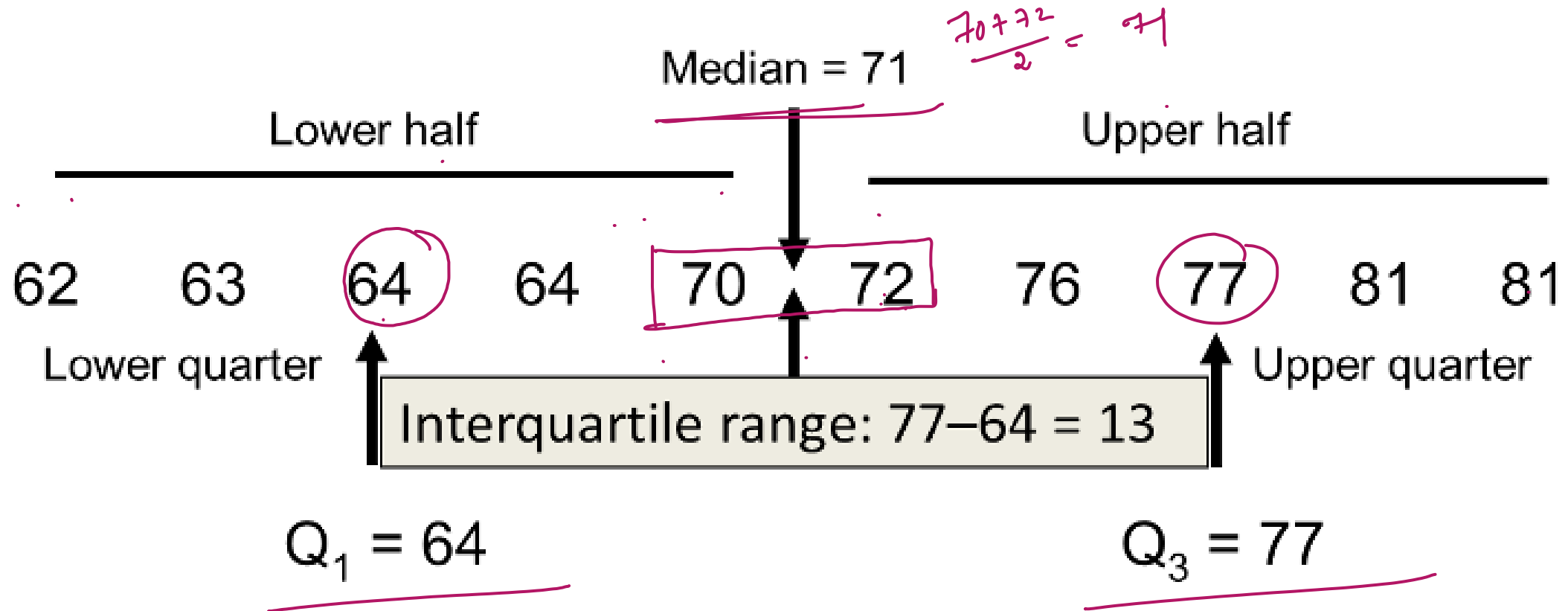
Q1 , Median (Q2) and Q3 (n is even)

$$\frac{\frac{n}{2} + (\frac{n}{2} + 1)}{2}$$

innovate

achieve

lead



Example:

Min = 5
Max = 30

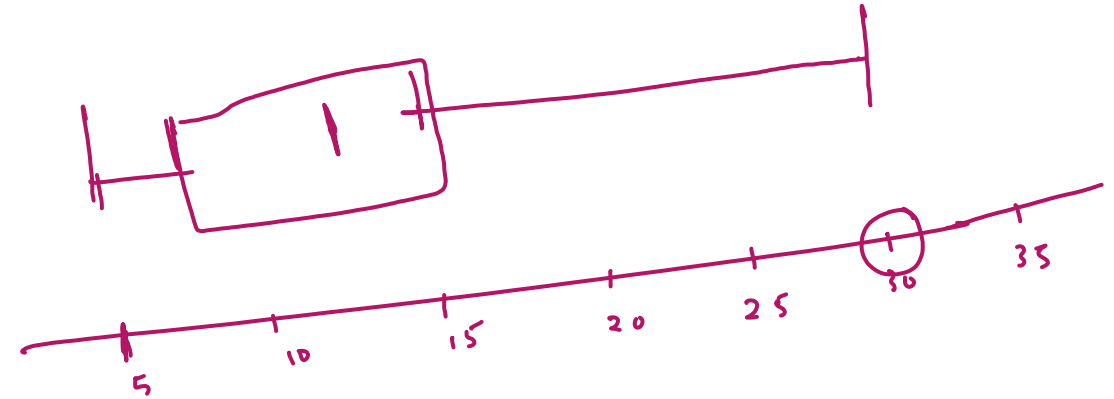


Data: 5, 7, 8, 9, 10, 12, 13, 15, 18, 30

Five-point summary:

- $Q1 = 8$ ✓
- $Q2$ (Median) = 11 ✓
- $Q3 = 15$ ✓
- $IQR = 15 - 8 = 7$
- Lower bound = $8 - (1.5 \times 7) = -2.5$ ✓
- Upper bound = $15 + (1.5 \times 7) = 25.5$ ✓

✓ Any value > 25.5 or < -2.5 is an **outlier** → here, **30** is an outlier.





- 2) Consider the following statistical summary of a dataset related to **student academic performance**.
Write at least three useful observations as part of **data pre-processing**.

	StudyHours	Attendance	TestScore	SleepHours	ScreenTime
count	500.00	500.00	500.00	500.00	500.00
mean	4.800000	82.500000	71.200000	6.700000	3.900000
std	2.100000	12.400000	15.600000	1.200000	2.000000
min	0.000000	35.000000	20.000000	3.000000	0.500000
25%	3.000000	75.000000	60.000000	6.000000	2.500000
50%	5.000000	84.000000	72.000000	7.000000	3.500000
75%	6.000000	91.000000	83.000000	7.500000	5.000000
max	12.000000	100.000000	99.000000	10.000000	12.000000

Questions

1. Identify possible outliers in the dataset.
2. Determine whether any variable contains suspicious or unrealistic values.
3. Comment on the spread and distribution of the variables using mean and standard deviation.

1. Study Hours: 12

Attendance: 35

Test Score: 20

Sleep Hours: 3 and 10

Screen Time: 12

Answers:



2. StudyHours = 0

A value of 0 study hours may indicate:

- missing data, ✓
- incorrect entry,
- or a genuine case. ✓

ScreenTime = 12 hours

Excessively high daily screen time may be unusual and should be verified.

Attendance = 35%

Very low attendance could indicate:

- abnormal behavior, ✓
- incomplete records, ✓
- or data entry issues. ✓

These values should be checked during preprocessing.

Answers:



3. StudyHours

Mean = 4.8 ✓

Std = 2.1 ✓

Moderate variation exists among students.

Attendance

Mean = 82.5 ✓

Std = 12.4 ✓

Attendance levels vary significantly.

TestScore

Mean = 71.2 ✓

Std = 15.6 ✓

Large spread indicates differences in student performance.

SleepHours

Mean = 6.7 ✓

Std = 1.2 ✓

Sleep duration is relatively consistent.

ScreenTime

Mean = 3.9

Std = 2.0

Considerable variation exists in screen usage habits.

$$(d) \& (e) \left\{ \begin{array}{l} A^c \cap B^c = (\overline{A \cup B}) \\ A^c \cup B^c = (\overline{A \cap B}) \end{array} \right\} \rightarrow \text{De Morgan's Laws}$$

Basic Probability

3) If $P(A) = 1/2$, $P(B) = 1/3$ and $P(A \cap B) = 1/5$ then find

a). $P(A \cup B)$

b). $P(A^c \cap B)$

c). $P(A \cap B^c)$

d). $P(A^c \cap B^c)$

e). $P(A^c \cup B^c)$

f). $P((A \cup B)^c)$

$$(a) P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{5} = \frac{15 + 10 - 6}{30} = \frac{19}{30}$$

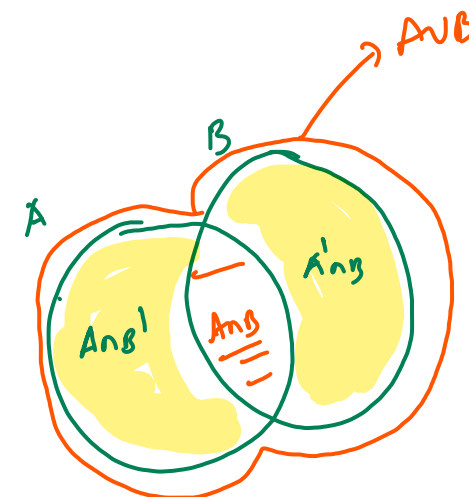
$$(b) P(A^c \cap B) = P(B) - P(A \cap B) = \frac{1}{3} - \frac{1}{5}$$

$$(c) P(A \cap B^c) = P(A) - P(A \cap B) = \frac{1}{2} - \frac{1}{5}$$

$$(d) P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 - \frac{19}{30}$$

$$(e) P(A^c \cup B^c) = P((A \cap B)^c) = 1 - P(A \cap B) = 1 - \frac{1}{5}$$

$$(f) P((A \cup B)^c) = 1 - P(A \cup B) = 1 - \frac{19}{30}$$



Solution

$$\text{a) } P(A \cup B) = P(A) + P(B) - P(A \cap B) = 1/2 + 1/3 - 1/5 = 19/30 = 0.6333$$

$$\text{b) } P(A^c \cap B) = P(B - A) = P(B - (A \cap B)) = P(B) - P(A \cap B) = 1/3 - 1/5 = 2/15 = 0.1333$$

$$\text{c) } P(A \cap B^c) = P(A - B) = P(A - (A \cap B)) = P(A) - P(A \cap B) = 1/2 - 1/5 = 3/10 = 0.3$$

$$\text{d) } P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 19/30 = 11/30 = 0.3667$$

$$\text{e) } P(A^c \cup B^c) = P((A \cap B)^c) = 1 - P(A \cap B) = 1 - 1/5 = 4/5 = 0.8$$

$$\text{f) } P((A \mid B)^c) = 1 - P(A \mid B) = 1 - 19/30 = 11/30 = 0.3667$$



Basic Probability

$$P(A \cap B) = P(A)P(B)$$
$$P(H \cap B_1 \cap B_2) = P(H)P(B_1)P(B_2) = (0.01)(0.02)(0.02)$$

4) There is a 1% probability for a hard drive to crash. Therefore, it has two backups, each having a 2 % probability to crash, and all three components are independent of each other. The stored information is lost only in an unfortunate situation when all three devices crash. What is the probability that the information is saved? ✓

Solution. Organize the data. Denote the events, say,

$H = \{ \text{hard drive crashes} \}$,

$B_1 = \{ \text{first backup crashes} \}$, $B_2 = \{ \text{second backup crashes} \}$.

It is given that H , B_1 , and B_2 are independent,

$$P\{H\} = 0.01, \text{ and } P\{B_1\} = P\{B_2\} = 0.02.$$

Applying rules for the complement and for the intersection of independent events,

$$\begin{aligned} P\{\text{saved}\} &= 1 - P\{\text{lost}\} = 1 - P\{H \cap B_1 \cap B_2\} \\ &= 1 - P\{H\}P\{B_1\}P\{B_2\} \\ &= 1 - (0.01)(0.02)(0.02) = 0.999996. \end{aligned}$$

Handwritten notes:

$$P(H) = 0.01$$
$$P(B_1) = 0.02$$
$$P(B_2) = 0.02$$
$$1 - P(\text{lost})$$
$$1 - 0.000004$$

*A → win A1
B → win A2*

Basic Probability

- 5) A political leader has submitted his nomination to compete in two different electoral constituencies namely A1 and A2. The probability of winning in constituency A1 and A2 is 0.80 and 0.65 respectively. The probability of losing at least one of the constituencies is 0.35. What will be the probability that he will win in exactly one of the constituencies?

*A^c - losing A1
B^c - losing A2*

We want:

*P(A) = 0.80
P(B) = 0.65
P(A^c ∪ B^c) = 0.35*

*⇒ P((A ∩ B)^c) = 0.35
⇒ 1 - P(A ∩ B) = 0.35
⇒ P(A ∩ B) = 1 - 0.35 = 0.65*

$P(\text{Win exactly one}) = P(\text{Win A1 and lose A2}) + P(\text{Lose A1 and win A2}) = 0.15 + 0 = 0.15$

- $P(\text{Win A1 and lose A2}) = P(A \cap B^c) = P(A) - P(A \cap B) = 0.80 - 0.65 = 0.15$
- $P(\text{Lose A1 and win A2}) = P(A^c \cap B) = P(B) - P(A \cap B) = 0.65 - 0.65 = 0$

Solution:

A political leader has submitted his nomination to compete in two different electoral constituencies namely A1 and A2. The probability of winning in constituency A1 and A2 is 0.80 and 0.65 respectively. The probability of losing at least one of the constituencies is 0.35. What will be the probability that he will win in exactly one of the constituencies?

Assume that A, B be the events defined as follows:

A : "Winning in constituency A1"

B : "Winning in constituency A2"

Given:

$$P(A) = 0.80, P(B) = 0.65$$

$$\text{and } P(\bar{A} \cup \bar{B}) = 0.35$$

Now, $\therefore P(\bar{A} \cup \bar{B}) = 0.35$

$$\therefore P(\overline{A \cap B}) = 0.35$$

$$\Rightarrow 1 - P(A \cap B) = 0.35$$

$$\Rightarrow P(A \cap B) = 0.65$$

Then, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= 0.80 + 0.65 - 0.65$$

$$\therefore P(A \cup B) = 0.80$$

$$\begin{aligned} \text{Then, } P(\text{He will win in one of the constituencies}) &= P(A \cup B) - P(A \cap B) \\ &= 0.80 - 0.65 \end{aligned}$$

$$\therefore P(\text{He will win in one of the constituencies}) = 0.15$$

$$\begin{aligned} P(\text{He will win in constituency A1 ONLY}) &= P(A) - P(A \cap B) \\ &= 0.80 - 0.65 \\ &= 0.15 \end{aligned}$$

$$\begin{aligned} P(\text{He will win in constituency A2 ONLY}) &= P(B) - P(A \cap B) \\ &= 0.65 - 0.65 \\ &= 0 \end{aligned}$$

$A \cup B = \{1, 2, 3, 4, 5, 6\}$
Exhaustive event

$S = \{1, 2, 3, 4, 5, 6\}$
 $A: \text{even no} \rightarrow \{2, 4, 6\}$
 $B: \text{odd no} \rightarrow \{1, 3, 5\}$

$A \cap B = \{ \}$

Mutually exclusive

Independent vs. Dependent Events



Using the bag of marbles on the left, what is the probability of pulling a black marble two times in a row? $P(\text{black, black})$

When you put 1st marble back in
(Independent Events)

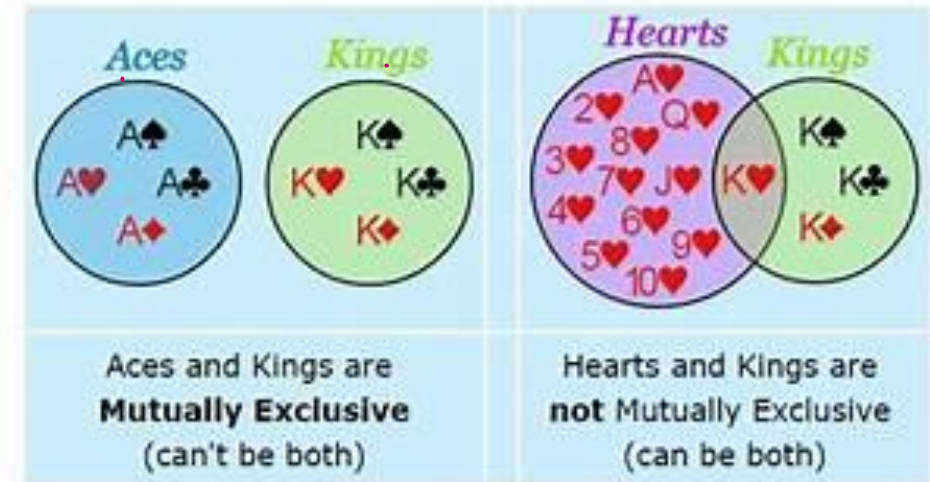
$$\frac{2}{10} * \frac{2}{10}$$

$$\frac{1}{5} * \frac{1}{5} = \frac{1}{25} \quad \checkmark$$

When you KEEP 1st marble
(Dependent Events)

$$\frac{2}{10} * \frac{1}{9}$$

$$\frac{1}{5} * \frac{1}{9} = \frac{1}{45}$$



Independent events

6) Comment on the statement:

“If two events are mutually exclusive, then they are independent also and vice versa”

$$\begin{aligned} \text{IE} &\rightarrow P(A \cap B) = P(A)P(B) \\ \text{ME} &\rightarrow P(A \cap B) = 0 \end{aligned}$$

Two independent events cannot be mutually exclusive events - unless one or both events have a probability of zero (meaning one of the events is impossible).

Independent events

7) Let A and B be the two possible outcomes of an experiment and suppose $P(A) = 0.4$, $P(B) = p$ and $P(A \cup B) = 0.7$

(i) For what choice of 'p' are A and B mutually exclusive?

(ii) For what choice of 'p' are A and B independent?

$$(i) P(A \cap B) = 0$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.7 = 0.4 + p - 0$$

$$\Rightarrow p = 0.7 - 0.4$$

$$p = 0.3$$

$$(ii) P(A \cap B) = P(A)P(B) = 0.4p$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.7 = 0.4 + p - 0.4p$$

$$0.7 - 0.4 = p(1 - 0.4)$$

$$0.3 = 0.6p$$

$$p = \frac{0.3}{0.6} = \frac{1}{2} = 0.5$$

$$\Rightarrow p = 0.5$$

Solution:

(i) If A and B are mutually exclusive then $P(A \cap B) = 0$

Thus $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ becomes

$$0.7 = 0.4 + P(B) - 0$$

$$\therefore P(B) = 0.7 - 0.4 = 0.3$$

(ii) If A and B are independent then $P(A \cap B) = P(A) \cdot P(B)$

Thus $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ becomes

$$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$$

$$0.7 = 0.4 + P - 0.4 \cdot P$$

$$= 0.4 + P(1 - 0.4)$$

$$= 0.4 + 0.6P$$

$$0.6P = 0.7 - 0.4 = 0.3$$

$$P = \frac{0.3}{0.6} = \frac{1}{2} = 0.5$$

$$\therefore P(B) = 0.5$$

Answer. (i) $p=0.3$, (ii) $p=0.5$



$$P(A \cap B) = P(A)P(B) \text{ (Independent)} \\ = P(A)P(B|A) \checkmark$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Conditional Probability

8. Ninety percent of flights depart on time. Eighty percent of flights arrive on time. Seventy-five percent of flights depart on time and arrive on time.

- a) You are meeting a flight that departed on time. What is the probability that it will arrive on time? $P(A|\underline{D}) = \frac{P(A \cap D)}{P(D)}$
- b) You have met a flight, and it arrived on time. What is the probability that it departed on time? $P(\underline{D}|A) = \frac{P(D \cap A)}{P(A)}$
- c) Are the events, departing on time and arriving on time, independent?

Let:

- D : Flight departs on time $0.9 \neq 0.72$
- A : Flight arrives on time Not
- $P(D) = 0.90 \checkmark$
- $P(A) = 0.80 \checkmark$
- $P(D \cap A) = 0.75 \checkmark$

$$P(A)P(D) = 0.9 \times 0.8 \\ = 0.72$$

$$P(A \cap D) = P(A)P(D) \rightarrow \text{they are independent}$$

$$P(A|\underline{D}) = \frac{P(A \cap D)}{P(D)} \\ = \frac{0.75}{0.90} \\ \downarrow \\ P(\underline{D}|A) = \frac{P(D \cap A)}{P(A)} \\ = \frac{0.75}{0.80}$$

Solution:



- a) You are meeting a flight that departed on time. What is the probability that it will arrive on time?

$$P(A | D) = \frac{P(D \cap A)}{P(D)}$$
$$P(A | D) = \frac{0.75}{0.90} = \frac{75}{90} = \frac{5}{6} \approx 0.833$$

- b) You have met a flight, and it arrived on time. What is the probability that it departed on time?

$$P(D | A) = \frac{P(D \cap A)}{P(A)}$$
$$P(D | A) = \frac{0.75}{0.80} = \frac{75}{80} = \frac{15}{16} = 0.9375$$

c) Are the events, departing on time and arriving on time, independent?

Two events A and B are independent if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Check:

$$P(D) \cdot P(A) = (0.90)(0.80) = 0.72$$

But:

$$P(D \cap A) = 0.75 \neq 0.72$$

So, the events are NOT independent.



Conditional Probability

$$\begin{aligned}P(F) &= 30\% = 0.30 \\P(A) &= 45\% = 0.45 \\P(F \cap A) &= 5\% = 0.05\end{aligned}$$

9) In an online shopping survey, 30% of persons made shopping in Flipkart, 45% of persons made shopping in Amazon and 5% made purchases in both. If a person is selected at random, find

- i) the probability that he makes shopping in at least one of two companies $P(F \cup A) = P(F) + P(A) - P(F \cap A)$
- ii) the probability that he makes shopping in Amazon given that he already made shopping in Flipkart. $P(A | F) = \frac{P(A \cap F)}{P(F)}$
- iii) the probability that the person will not make shopping in Flipkart given that he already made purchase in Amazon.

$$P(F' | A) = \frac{P(F' \cap A)}{P(A)} = \frac{P(A) - P(F \cap A)}{P(A)}$$

Conditional Probability

Solution: Given $P(F) = 30\% = 0.30$

$$P(A) = 45\% = 0.45$$

$$P(F \cap A) = 5\% = 0.05$$

$$\text{i) } P(F \cup A) = P(F) + P(A) - P(F \cap A)$$

$$= 0.30 + 0.45 - 0.05 = 0.7$$

$$\begin{aligned} \text{ii) } P(A | F) &= \frac{P(A \cap F)}{P(F)} \\ &= \frac{0.05}{0.30} = 0.167 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(F' | A) &= \frac{P(F' \cap A)}{P(A)} \\ P(F' \cap A) &= P(A) - P(A \cap F) \\ &= 0.45 - 0.05 \\ &= 0.40 \end{aligned}$$

$$P(F' | A) = \frac{0.40}{0.45} = 0.88$$

$$P(s) = P(o \cap s) + P(B \cap s) + P(P \cap s)$$

Total Probability

$$P(s) = P(o) \cdot P(s|o) + P(P) \cdot P(s|P) + P(B) \cdot P(s|B)$$

office $\rightarrow$ span
 perm $\rightarrow$ span
 B25 $\rightarrow$ sic



10) A businessman goes to hotels X, Y, Z -20%, 50%, 30% of the time, respectively. It is known that 5%, 4%, 8% of the rooms in X, Y, Z hotels have faulty plumbing. Determine the probability that the businessman goes to hotel with faulty plumbing.

Given:

Hotel Visit Probabilities:

- $P(X) = 0.20$
- $P(Y) = 0.50$
- $P(Z) = 0.30$

Probability of Faulty Plumbing (given hotel):

- $P(F|X) = 0.05$
- $P(F|Y) = 0.04$
- $P(F|Z) = 0.08$

Find the **total probability** that the businessman goes to a hotel with faulty plumbing:

$$P(F) = P(X) \cdot P(F|X) + P(Y) \cdot P(F|Y) + P(Z) \cdot P(F|Z)$$

$$P(F) = (0.20 \times 0.05) + (0.50 \times 0.04) + (0.30 \times 0.08)$$

$$= 0.010 + 0.020 + 0.024 = \boxed{0.054}$$

Total Probability

11) A company receives 50% of customer complaints via phone, 30% via email, and 20% via chat.

Resolution rates:

- Phone: 90%
- Email: 80%
- Chat: 70%

What is the probability that a randomly selected complaint is resolved?

12
Given:

- $P(\text{Phone}) = 0.50$ (50% of complaints come via phone),
- $P(\text{Email}) = 0.30$ (30% of complaints come via email),
- $P(\text{Chat}) = 0.20$ (20% of complaints come via chat),
- $P(\text{Resolved} \mid \text{Phone}) = 0.90$ (90% of phone complaints are resolved),
- $P(\text{Resolved} \mid \text{Email}) = 0.80$ (80% of email complaints are resolved),
- $P(\text{Resolved} \mid \text{Chat}) = 0.70$ (70% of chat complaints are resolved).

To calculate $P(\text{Resolved})$, we'll use the **Law of Total Probability**. The formula is:

$$P(\text{Resolved}) = P(\text{Phone}) \cdot P(\text{Resolved} \mid \text{Phone}) + P(\text{Email}) \cdot P(\text{Resolved} \mid \text{Email}) + P(\text{Chat}) \cdot P(\text{Resolved} \mid \text{Chat})$$

Solution:

$$P(\text{Resolved}) = 0.50 \cdot 0.90 + 0.30 \cdot 0.80 + 0.20 \cdot 0.70 = 0.45 + 0.24 + 0.14 = 0.83$$

Conclusion:

So, there is an **83%** chance that a randomly selected customer complaint will be resolved, based on the given resolution rates for each channel.

Problem:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



12. In a factory, three machines, J , K and L , are used to make biscuits.

Machine J makes 25% of the biscuits.

Machine K makes 45% of the biscuits.

The rest of the biscuits are made by machine L . $\rightarrow 30\%$

It is known that 2% of the biscuits made by machine J are broken, 3% of the biscuits made by machine K are broken and 5% of the biscuits made by machine L are broken.

- (a) Draw a tree diagram to illustrate all the possible outcomes and associated probabilities. (2)

A biscuit is selected at random.

- (b) Calculate the probability that the biscuit is made by machine J and is not broken. (2)
- (c) Calculate the probability that the biscuit is broken. (2)
- (d) Given that the biscuit is broken, find the probability that it was not made by machine K . (3)

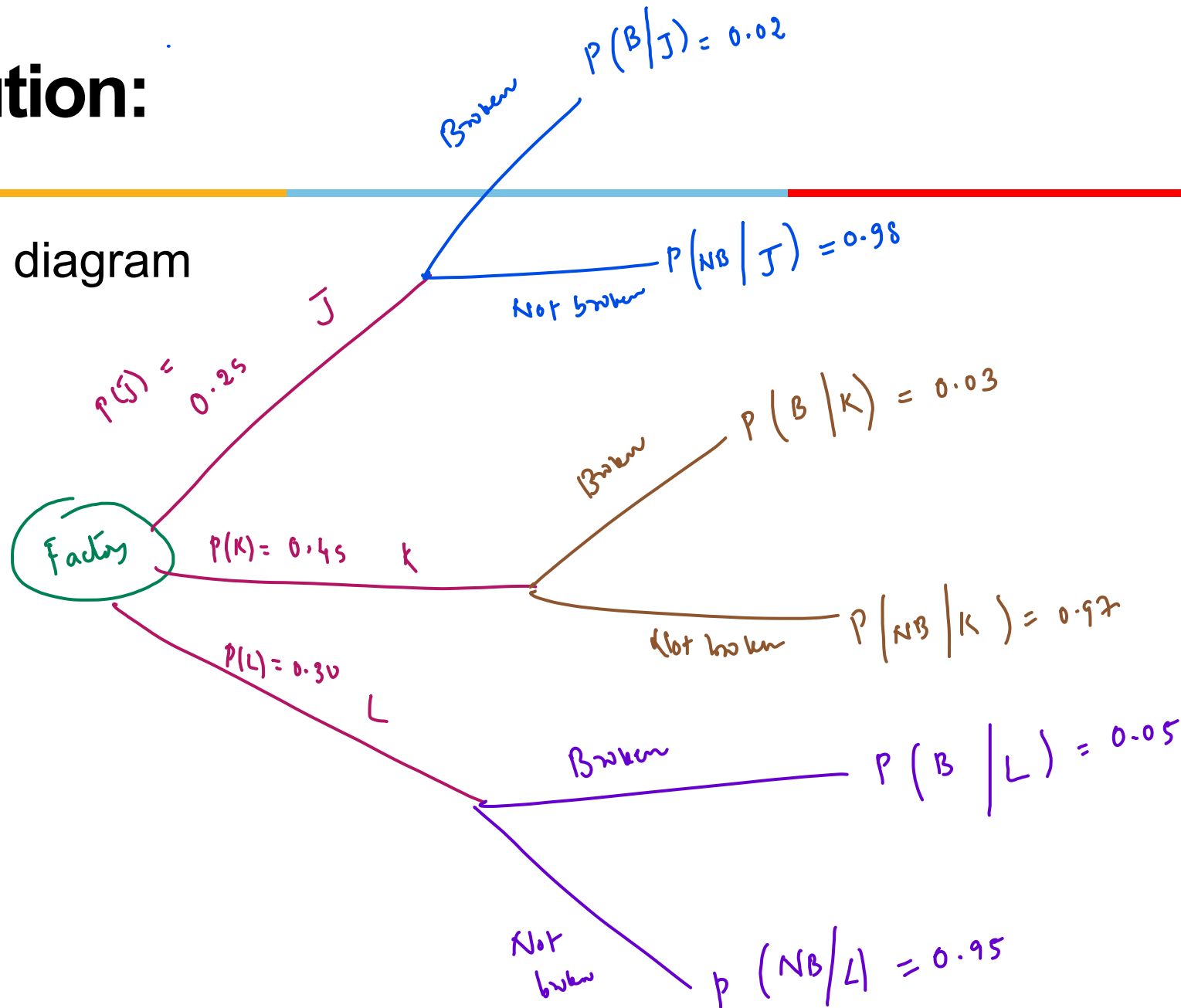
$$\begin{aligned} P(\text{Not } K \cap \text{Broken}) &= P(J \cap B) + P(K \cap B) \\ &= (0.25)(0.02) + (0.45)(0.03) \\ &= 0.020 \end{aligned}$$

$$\begin{aligned} P(J \cap NB) &= P(J) \cdot P(NB|J) \\ &= (0.25)(0.98) \\ P(\text{Not } K | \text{Broken}) &= \frac{P(\text{Not } K \cap \text{Broken})}{P(\text{Broken})} \\ &= \frac{0.020}{0.0335} = 0.597 \end{aligned}$$

Solution:



a) Tree diagram



$$\begin{aligned} P(B) &= P(J \cap B) + P(K \cap B) + P(L \cap B) \\ &= P(J) \cdot P(B|J) + P(K) \cdot P(B|K) + P(L) \cdot P(B|L) \\ &= (0.25)(0.02) + (0.45)(0.03) + (0.30)(0.05) \\ &= 0.0335 \end{aligned}$$

Solution:



(b) Probability biscuit is made by J and not broken

$$\begin{aligned}P(J \text{ and not broken}) &= P(J) \cdot P(\text{not broken} | J) \\&= 0.25 \cdot 0.98 = 0.245\end{aligned}$$

So, the probability is **0.245 (24.5%)**.

(c) Probability biscuit is broken

We use the law of total probability:

$$\begin{aligned}P(\text{broken}) &= P(J) \cdot P(\text{broken}|J) + P(K) \cdot P(\text{broken}|K) + P(L) \cdot P(\text{broken}|L) \\&= (0.25 \cdot 0.02) + (0.45 \cdot 0.03) + (0.30 \cdot 0.05) \\&= 0.005 + 0.0135 + 0.015 = 0.0335\end{aligned}$$

So, the probability is **0.0335 (3.35%)**.

d) Given biscuit is broken, probability it was not made by K

We use conditional probability:

$$P(\text{not } K / \text{broken}) = \frac{P(\text{broken and not } K)}{P(\text{broken})}$$

$$\begin{aligned} P(\text{broken and not } K) &= P(\text{broken and } J) + P(\text{broken and } L) \\ &= (0.25 \cdot 0.02) + (0.30 \cdot 0.05) = 0.005 + 0.015 = 0.020 \end{aligned}$$

$$P(\text{not } K / \text{broken}) = \frac{0.020}{0.0335} \approx 0.597$$

So, the probability is about **0.597 (59.7%)**.

Problem:

$$P(A \cup B \cup C) = P(S)$$

$$P(A) + P(B) + P(C) = 1$$

innovate

achieve

lead

13. Suppose A, B and C are three mutually exclusive events in a sample space. Given that $S = A \cup B \cup C$, $P(A) = (1/5)P(B)$, and $P(C) = 4P(A)$, find $P(B \cup C)$.

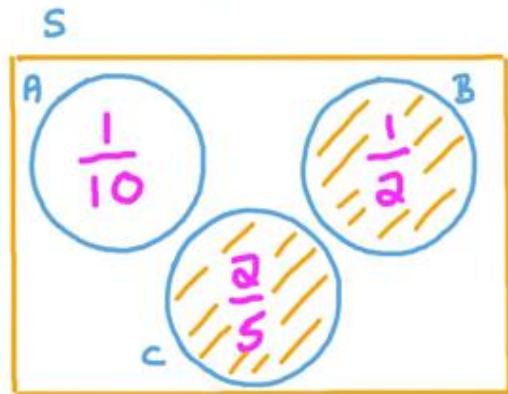
Solution:

$$= P(B) + P(C) - P(B \cap C)$$

$$= P(B) + P(C)$$

$$5P(A) = P(B)$$

Two or more events are mutually exclusive if they cannot happen at the same time. $P(X \cup Y) = P(X) + P(Y)$



$$P(A \cup B \cup C) = 1$$

$$P(A) + P(B) + P(C) = 1$$

$$P(A) + 5P(A) + 4P(A) = 1$$

$$10P(A) = 1$$

$$P(A) = \frac{1}{10}$$

$$P(B) = \frac{5}{10} = \frac{1}{2}$$

$$P(C) = \frac{4}{10} = \frac{2}{5}$$

$$P(B \cup C) = P(B) + P(C)$$

$$= \frac{5}{10} + \frac{4}{10}$$

$$= \frac{9}{10}$$

Practice Problems

1. If $P(A) = 1/3$, $P(B) = 1/2$, $P(A/B) = 1/6$ find i). $P(B/A)$ ii). $P(B/A')$ iii). $P(A \cup B / A)$ iv). $P(B/A)$.

2. Three machines A, B, C produce 50%, 30%, and 20% of the items in a factory. The percentage of defective outputs of these machines are 3, 4 and 5 respectively.
 - (i) If an item is selected at random, what is the probability that it is defective? (Ans: 0.037)
 - (ii) If a selected item is defective, what is the probability that it is from machine A? (Ans: 0.4054)



Practice Problems

3. A weather app predicts rain with 90% accuracy when it actually rains. It wrongly predicts rain 10% of the time when it doesn't rain. Suppose it rains only 10% of days in a year. You check the app and it predicts rain. What is the probability it will actually rain? (Ans: 0.5)

4. Consider the following data related to the employees, who are on travel. 40% check work email, 20% use cell phone to stay connected to work, 25% bring laptop with them, 23% check both work email and use cell phone to stay connected, and 50% neither check work email nor use a cell phone to stay connected nor bring a laptop. In addition, 88 out of every 100 who bring a laptop also check work email, and 70 out of every 100 who use a cell phone to stay connected also bring a laptop.

- i) What is the probability that a randomly selected traveller who checks work email also uses a cell phone to stay connected?
- ii) What is the probability that someone who brings a laptop on vacation also uses a cell phone to stay connected?
- iii) If the randomly selected traveller checked work email and brought a laptop, what is the probability that he/she uses a cell phone to stay connected?

TEXT BOOKS

T1 : Statistics for Data Scientists, An introduction to Probability,
Statistics and Data Analysis, Maurits Kaptein et al, Springer 2022

T2 : Probability and Statistics for Engineering and Sciences,
8th Edition, Jay L Devore, Cengage Learning

T3 : Introduction to Time Series and Forecasting, Second Edition,
Peter J Brockwell, Richard A Davis, Springer.

Questions please...



THANK YOU!